Domain Adaptation for Remote Sensing Image Semantic Segmentation: An Integrated Approach of Contrastive Learning and Adversarial Learning

Lubin Bai, Shihong Du, Xiuyuan Zhang, Haoyu Wang, Bo Liu, and Song Ouyang

***Abstract*— Although semantic segmentation models based on deep neural networks (DNNs) have achieved excellent results, generalizing well from one remote sensing dataset (source domain) to another dataset with different acquisition conditions (target domain) remains a major challenge. Many domain adaptation (DA) approaches have been proposed to address this problem. DA aims to help DNNs learn a generalizable representation space in which source and target domains have similar feature distributions, but most of the existing DA approaches have difficulty in aligning the high-dimensional image representations of two domains directly. In this study, we proposed a model integrating contrastive learning and adversarial learning in a unified framework for aligning two domains in both representation space and spatial layout. Specifically, the model consists of a semantic segmentation network for feature extraction and two branches for DA. The first branch is used for adaptation in representation space directly by a proposed pixelwise contrastive loss, while the second branch is used for adaptation in predicted results to help two domains have similar spatial layouts through a novel but simple entropy-based similarity discriminator. Additionally, a training strategy called category similarity matching sampling was proposed to provide source and target image pairs with similar category composition for each training iteration, which can help the two branches work better. Extensive experiments indicated that the two branches can benefit each other to gain a superior performance and DA pretraining by our methods can achieve impressive results with only a small number of target labeled samples.**

***Index Terms*— Adversarial learning, contrastive learning, domain adaptation (DA), semantic segmentation.**

I. Introduction

DURING the last 20 years, with the successful launch of a large number of remote sensing (RS) satellites, an increasing number and variety of RS images [1] have been acquired. However, in contrast to the rapidly increased ability of acquiring RS image data, the current automatic interpretation ability of RS images is still very low. Most existing RS image interpretation methods require a huge number of manually annotated sample data to achieve satisfactory results, which consumes a lot of time and labor [2], [3]. In addition, a model trained on some labeled RS images (source domain) usually suffers a radical drop in performance when applied to other images with different acquisition conditions (target domain), such as different ground sampling distances (GSDs), different spectral bands, and different regions. The reason lies in the great variations in data distributions in two domains, typically known as domain shift [4], [5].

Semantic segmentation for RS images, as an important dense pixel classification task, is the foundation for land-cover and land-use classifications. It has been extensively used for urban planning [6], [7], change detection [8], environment monitoring [9], and so on. Recently, deep neural networks (DNNs) have achieved inspiring success in semantic segmentation. However, most of these methods highly depend on strong supervision (large quantity of well-annotated pixel-level labeled data to train the models) and, thus, face the challenge of domain shift. To address this issue, domain adaptation (DA) approaches have been proposed to bridge the gap between source and target domains. Mimicking the human vision system, DA is a particular case of transfer learning (TL) that aims at transferring knowledge from the source domain to the target domain. For DNNs-based semantic segmentation models, DA aims to help them learn a generalizable feature space, in which source and target domains have similar feature distributions. Adversarial learning is a common method used for semantic segmentation DA problem [10], but most of them have difficulty in handling high-dimensional features directly. Contrastive learning can contrast the features from different distributions and has achieved extraordinary results in unsupervised image feature representation [11], [12], but in the context of semantic segmentation DA problem, it remains largely underexplored. The above two methods shed some light on solving the DA problem of RS image semantic segmentation, and thus, we will then review the related work.

Adversarial learning for DA is the most explored approach for semantic segmentation [13], which usually involves two networks. One is the segmentation network that takes the source and target domain images as input, while another is the discriminator network that takes the outputs from the segmentation network as input and tries to predict which domain it comes from. The segmentation network tries to fool the discriminator, thus making the outputs from two domains have a similar spatial layout. Hoffman *et al.* [14] are the first to introduce the task of DA into semantic segmentation by applying adversarial training in a fully convolutional way on feature

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representations. However, high-dimensional image features often contain noise, which can interfere with the training of the discriminator. Since semantic segmentation is a pixel-level prediction task and produces structured outputs with spatial and local information, Tsai *et al.* [15] proposed an efficient DA algorithm through adversarial training on the predicted labels. After that, some new variants of the DA model based on adversarial training were proposed, such as category-level adversarial network (CLAN) [16] and fine-grained adversarial approach (FADA) [17]. Both emphasize the importance of category-level feature alignment in reducing domain shift. The former tries to adaptively weight the adversarial loss for different features, while the latter represents the supervision of traditional discriminator at a fine-grained semantic level. Based on the DA models in the computer vision field, some DA approaches for RS image semantic segmentation have been proposed. Yan *et al.* [18] designed a triplet adversarial domain adaptation (TriADA) model, which takes both domains into account to learn a domain-invariant classifier for RS image semantic segmentation. Luo *et al.* [16] proposed a generative approach for RS image semantic segmentation, which used dual generative adversarial network (DualGAN) to generate simulated samples like the target images and then employed multiple weakly supervised methods to align target and source domains. The adversarial training on the predicted labels can take the spatial structure of an image into account, but it often faces the problem of training stability and is not suitable for dealing with high-dimensional and complex features in semantic segmentation directly [15].

Unlike adversarial training, the idea of contrastive learning is similar in spirit to the soft nearest-neighbor loss [19], [20], which encourages soft feature alignments and is more robust to potential noise in the features [21]. Contrastive learning aims to learn a representation space by contrasting semantically positive and negative pairs of samples [12], [22]. Several recent studies [12], [23], [24], [25] present promising results on unsupervised visual representation learning using approaches related to the contrastive loss. Recently, some works have shown that for semantic segmentation task, which requires high-dimensional features at the pixel level, contrastive learning can also help to give a proper representation to the pixels in images [26], [27], [28], [29]. However, these works focus on self-supervised contrastive learning aiming to learn a feature representation for downstream tasks, whose idea and design are different from DA. For semantic segmentation DA problem, Liu *et al.* [30] proposed a method via pathwise contrastive learning. They divide an image into patches at several resolution levels and use contrastive learning to make the representations of similar patches close to each other. Compared with adversarial learning, there are fewer studies using contrastive learning in RS image semantic segmentation DA problem currently, and we will utilize its powerful feature extraction capability for the source and target domain alignment problems.

Most adversarial training methods for semantic segmentation DA are based on the predicted results, which are spatially structured outputs, and adversarial training on the high-dimensional features directly usually suffers from noise. On the other hand, contrastive learning allows for soft alignment in the feature space, but it lacks the ability to represent the spatial structure of an image. Considering the above reasons, we propose an end-to-end model that considers both representation space and spatial structure by combining contrastive learning and adversarial learning. Our model includes a semantic segmentation network and two branches. The segmentation network is used for extracting features and making predictions on the source and target images, while the two branches work together to help to align two domains. The first branch is used for adaptation in representation space (ADRS) directly based on a proposed pixelwise contrastive loss. The proposed contrastive loss models pixel relationships in the representation space directly, which encourages the pixel features of the same class in source and target domains to be similar and pixel features of different classes to be dissimilar, and in this way, the segmentation network can extract domain-invariant features for two domains. The second branch is used for adaptation in predicted results (ADPR) between two domains. This branch employs a novel but simple entropy-based similarity discriminator. We alternately optimize the discriminator and the segmentation network by playing a minimax game as GANs do. Since the discriminator can see the prediction of a complete image, it can help the segmentation network to learn spatial information. In addition, we also propose a training strategy called category similarity matching sampling (CSMS) to provide source and target image pairs with similar category composition for each training iteration. This strategy can provide proper positive and negative samples for contrastive loss in training, as well as improve the discriminator's ability to distinguish between similar source and target images.

In summary, the main contributions of this work can be summarized as follows.

A. New Loss Function

The previous contrastive loss functions are not suitable for the semantic segmentation DA problem. Therefore, we extend contrastive learning to the task of semantic segmentation DA and propose a new pixelwise contrastive loss.

B. New Framework and a New Training Strategy

An end-to-end model framework based on contrastive learning and adversarial learning for RS image semantic segmentation DA and a new training strategy are proposed. The model can simultaneously align source and target domains in terms of feature representations and spatial structures. In addition, this framework can be easily adopted by most of the existing semantic segmentation networks for the DA task.

II. Methodology

A. Overview

We focus on the DA problem for RS image semantic segmentation, where we have access to the source dataset $S = \{(x_s^1, y_s^1), (x_s^2, y_s^2), \ldots, (x_s^m, y_s^m)\}$ containing m images along with associated C-class segmentation labels, and the target dataset $T = \{x_t^1, x_t^2, \ldots, x_t^n\}$ containing n images without

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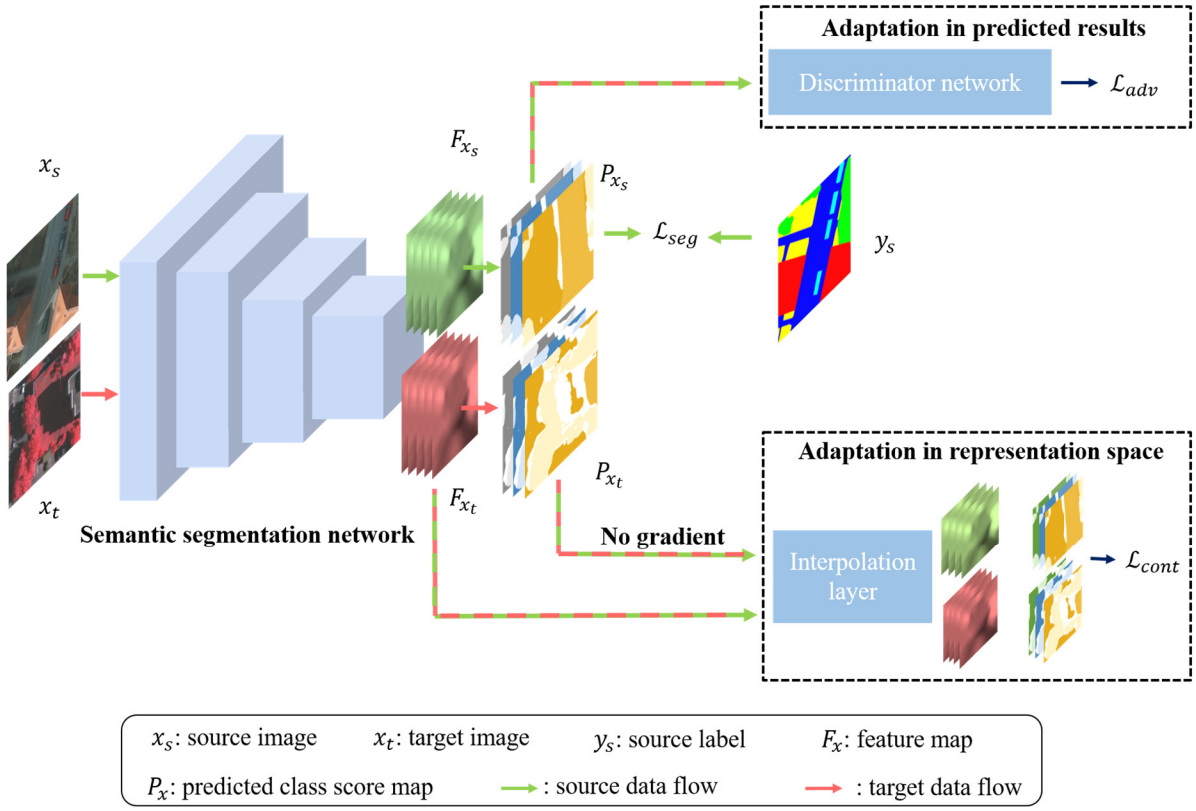


Fig. 1. Flowchart of the proposed model. The framework includes three modules: semantic segmentation network, ADRS branch, and ADPR branch.

labels. S and T can be RS images from different locations, different times, different sensors, and so on. Our goal is to train a segmentation model G with an accurate prediction in the target domain by using datasets S and T. Therefore, the DA method should train G that can distill knowledge from labeled source data and learn domain-invariant features. As shown in Fig. 1, in this work, we aim at developing a generic end-to-end model for RS image segmentation DA. To build our model, we start from the existing semantic segmentation networks and add two branches for DA. For each training iteration, we need to input one source image and one target image, and their feature maps and predicted results will be produced by the segmentation model G. The ADRS branch takes feature maps and predicted results as input and uses them to calculate the pixelwise contrastive loss of source and target domains. The ADPR branch employs a novel but simple entropy-based similarity discriminator D similar to the one in [13]. D takes in the weighted self-information maps (we will describe it in Section II-D) of G's predicted score map and calculates the adversarial loss. These two branches can complement each other and give an impressive result on the target data. The final loss of the semantic segmentation network consists of three parts

$$\mathcal{L} = (1 - \lambda_1 - \lambda_2)\mathcal{L}_{\text{seg}} + \lambda_1 \mathcal{L}_{\text{cont}} + \lambda_2 \mathcal{L}_{\text{adv}} \tag{1}$$

where $\mathcal{L}_{\text{seg}}$ is the cross-entropy loss of source images, $\mathcal{L}_{\text{cont}}$ is the contrastive loss calculated by ADRS branch, $\mathcal{L}_{\text{adv}}$ is the adversarial loss calculated by ADPR branch, and λ_1 and λ_2 are the weights of $\mathcal{L}_{\text{cont}}$ and $\mathcal{L}_{\text{adv}}$, respectively. The aim of $\mathcal{L}_{\text{seg}}$ is to distill knowledge from labeled source data, while $\mathcal{L}_{\text{cont}}$ and $\mathcal{L}_{\text{adv}}$ can help to learn domain-invariant features. In addition, we propose a training strategy to help the two branches work better.

B. Semantic Segmentation Network

The segmentation network G is used for feature extraction and result prediction. The ADRS branch and ADPR branch for DA have no special requirements for G; hence, most of the existing popular semantic segmentation networks can be used in our framework. Considering the model size and the ability to extract features, in this work, DeepLabv3+ is adopted. The outputs of G consist of two parts: a feature map F and a predicted class score map P produced by the classifier based on F. According to the structure of DeepLabv3+ [31], F is the concatenation features of the high-level and low-level features from the network. During training, each time we sample an x_{s} and x_{t} from S and T, and feed them into the segmentation network separately, and then, the network will output a feature map and a predicted class score map for each input, which are denoted as $F_{x_{\text{s}}}$, $P_{x_{\text{s}}}$, $F_{x_{\text{t}}}$, and $P_{x_{\text{t}}}$. After that we calculate the multiclass cross-entropy loss on the source domain data

$$\mathcal{L}_{\text{seg}} = -\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} y_{\text{s}}^{(h,w,c)} \log\left(P_{x_{\text{s}}}^{(h,w,c)}\right) \tag{2}$$

where H, W, and C denote the height, width, and number of classes of the source image, respectively.

We can improve the performance of the model on the source dataset with segmentation loss. However, to improve

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the prediction on the target dataset, we need to introduce the ADRS and ADPR modules for DA.

C. Adaptation in Representation Space

For training the segmentation network using L_{seg} alone, the features extracted by the model on source and target domains will have a large gap, and thus, we use the ADRS branch to align them directly. Contrastive loss measures the similarities of sample pairs in representation space. ADRS branch employs a contrastive loss to conduct training by contrasting the representations of source and target images. In this way, representations of the same class in two domains are similar, and thus, the classifier can achieve a good result on the target dataset. As noted in [12], a key to contrastive learning is to provide the positive samples with sufficient negative samples for contrast. Therefore, in our study, we treat each pixel in one image as a sample, so that a single image contains enough samples for contrastive learning. The next difficulty lies in selecting the positive and negative samples. Existing work on self-supervised contrastive learning circumvents the problem by forcing the positive examples to come from another augmentation of the same image, whether using a large mini-batch [22] or a memory bank [25] as the source of negative examples. These methods aim to teach the model to recognize the same image from different augmented views, so that it can extract good features helpful for image interpretation. Unlike self-supervised learning, as shown in Fig. 2, in DA problem, target and source domains have inherently large gaps in the original representation space, and our purpose is to reduce the gap between them, in other words, to make their representation distributions similar. What we need is to learn a joint embedding space where the pixel features of the same class in source and target domains are similar and pixel features of the different classes are dissimilar. To achieve this goal, we require the class information to construct contrastive samples. Specifically, for class $\overline{c}$ in the target domain, we can take class $\overline{c}$ in the source domain as positive and the other classes in the source domain as negative, and then, we use the contrastive loss to force the class $\overline{c}$ in the target domain closer to the $\overline{c}$ in the source domain. Unlike supervised contrastive learning [32] which uses image labels to select positive and negative samples, in DA problem, labels of the target dataset are unreachable. Consequently, there are two options providing class information for us to construct contrastive samples: 1) using ground truths of the source domain and predicted labels of the target domain and 2) using predicted labels of two domains. We choose option (2) because it gives better results, and we will explain the reasons for this further in the experiments in Section III-D.

Based on the above analysis, we extend the contrastive loss to the DA for semantic segmentation. Our contrastive loss can align source and target domains directly using the feature map extracted by the semantic segmentation network. Each time we sample a source image x_{s} and a target image x_{t} from S and T, and input them into the segmentation network, which will output their feature maps $F_{x_{\text{s}}}$ and $F_{x_{\text{t}}}$. For a pixel i_{t} from x_{t} with its predicted label $\overline{c}$, the positive samples are pixels from $F_{x_{\text{s}}}$ that are predicted as the same class $\overline{c}$, while the

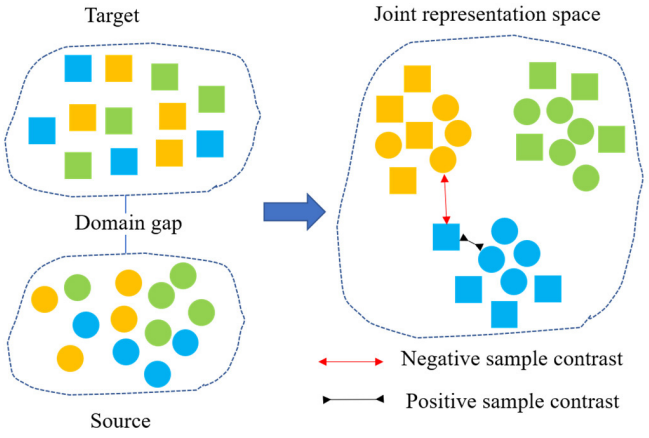


Fig. 2. Visualization of the proposed contrastive loss.

negative samples are pixels that come from x_{s} and predicted as the other classes $C\backslash\overline{c}$. Our pixelwise contrastive loss for i_{t} is defined as

$$\mathcal{L}_{\text{cont}}^{i_{\text{t}}} = \frac{1}{|F_{x_{\text{s}}}^{i_{\text{t}}+}|}\sum\nolimits_{\vec{j}\in F_{x_{\text{s}}}^{i_{\text{t}}+}} -\log \times \frac{\exp\left(\vec{i}_t\cdot\vec{j}/\tau\right)}{\exp\left(\vec{i}_t\cdot\vec{j}/\tau\right)+\frac{1}{|F_{x_{\text{s}}}^{i_{\text{t}}-}|}\sum_{\vec{z}\epsilon F_{x_{\text{s}}}^{i_{\text{t}}-}}\exp\left(\vec{i}_t\cdot\vec{z}/\tau\right)} \tag{3}$$

where $\vec{i}_t$ is the corresponding feature for i_{t} from $F_{x_{\text{t}}}$, $F_{x_{\text{s}}}^{i_{\text{t}}+}$ and $F_{x_{\text{s}}}^{i_{\text{t}}-}$ denote the positive and negative pixel feature collections for i_{t}, respectively, $|F_{x_{\text{s}}}^{i_{\text{t}}+}|$ and $|F_{x_{\text{s}}}^{i_{\text{t}}-}|$ stand for the number of pixels in $F_{x_{\text{s}}}^{i_{\text{t}}+}$ and $F_{x_{\text{s}}}^{i_{\text{t}}-}$, respectively, $\vec{j}$ is a positive pixel feature in $F_{x_{\text{s}}}^{i_{\text{t}}+}$, $\vec{z}$ is a negative pixel feature in $F_{x_{\text{s}}}^{-}(i_{\text{t}})$, and $\tau > 0$ is a temperature hyperparameter. All pixel features are normalized before calculation. In (3), it is worth noting that our contrastive loss calculates the average of all the negative sample features in the denominator. There are two main reasons. First, the number of negative samples varies from image to image, and using the sum of negative samples as InfoNCE [33] does would lead to unstable values of the denominator. Averaging can ensure that the value of the denominator is relatively stable. Second, averaging is equivalent to replacing all $|N(i)|$ negative samples with one representative negative sample, reducing an $|N(i)|+1$ classification problem to a binary classification problem. For all pixels in x_{t}, we have

$$\mathcal{L}_{\text{cont}} = \frac{1}{|x_{\text{t}}|}\sum\nolimits_{i_{\text{t}}\in x_{\text{t}}}\mathcal{L}_{\text{cont}}^{i_{\text{t}}} \tag{4}$$

where $|x_{\text{t}}|$ means the number of pixels in x_{t}. From (3) and (4), we can see that the memory complexity of this loss is quadratic in the number of pixels. To avoid GPU memory issues as shown in Fig. 1, we resize the feature map using bilinear resizing in the ADRS branch before computing the contrastive loss.

D. Adaptation in Predicted Results

Although the ADRS branch can align the source and target domains in the representation space, it ignores the spatial structure information of an image. To compensate for this,

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in the ADPR branch, we utilize an adversarial method based on the weighted self-information proposed in [13] to take the structural dependencies between local semantics into account. Compared to other adversarial training methods, this method aims at not only entropy minimization in the target domain but also structure adaptation from the source domain to the target domain.

In detail, given a predicted class score map $P_x \in \mathbb{R}^{H\times W\times C}$ generated by the segmentation network, the self-information is defined as $-\log P_x$ [13]. We implement DA by adversarial training on weighted self-information map $I_x \in \mathbb{R}^{H\times W\times C}$, where $I_x = -P_x \log P_x$. It can be seen that the sum of the weighted self-information vector of each pixel is the Shannon entropy [34] of that pixel. For the discriminator D, we use a small fully convolutional network with parameter θ_D which takes I_x as input and produces domain classification outputs. We train the discriminator to discriminate the I_x coming from source and target images using a cross-entropy domain classification loss L_{CE}. At the same time, the segmentation network is trained to fool the discriminator. For discriminator, the training loss of the input I_{x_s} and I_{x_t} is

$$\mathcal{L}_D = \mathcal{L}_{\text{CE}}(D(I_{x_\text{s}}), 1) + \mathcal{L}_{\text{CE}}(D(I_{x_\text{t}}), 0). \tag{5}$$

The adversarial loss of the input x_t to train the segmentation network is

$$\mathcal{L}_{\text{adv}} = \mathcal{L}_{\text{CE}}(D(I_{x_\text{t}}), 1) \tag{6}$$

$\mathcal{L}_D$ optimizes the parameters of the discriminator network D, while $\mathcal{L}_{\text{adv}}$ optimizes the parameters of the segmentation network G. With the help of the ADPR branch, source and target domains will share similarities in the semantic layout. In addition, by combining the ADRS and ADPR branches, better results can be obtained, which will be shown in the experiments.

E. CSMS Strategy

During model training, each time we need to input a source image x_s and a target image x_t to the model. The traditional approach is to select a random image from each of the two datasets to form a pair. However, there is aggregation in the distribution of land-cover categories [35] in RS images, which can, therefore, lead to the unbalanced distribution of categories among image patches. Matching the same target image x_t to different source domain images brings different results. As shown in Fig. 3, the main land cover in x_s^1 is low vegetation and trees, whereas in x_t^1, they are mainly buildings, cars, and impervious surfaces, with a small proportion of low vegetation and trees. If we match x_t^1 and x_s^1 as a pair to input to the model, it would bring little benefit to the model, because in the ADPR branch, the discriminator can easily distinguish them as the similarity between x_t^1 and x_s^1 is very low and in the ADRS branch, there are very few positive pixels in x_s^1 when we calculate the contrastive loss. On the contrary, if we match x_t^1 and x_s^2 as a pair, the gain to the model would be satisfying. For the above reasons, we propose a training strategy, namely, category similarity matching, which can match the source and target images with similar land-cover together.

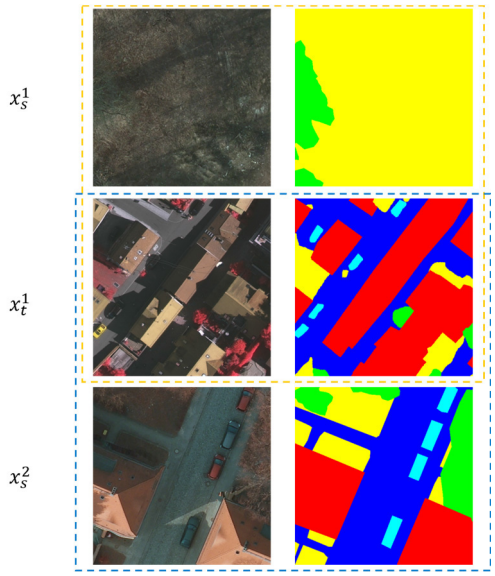


Fig. 3. Imbalance category distributions between RS image patches.

During the training process, we use the predicted results of the semantic segmentation model to calculate the similarity of the images. In detail, $R_x \in \mathbb{R}^{H\times W}$ is the predicted labels of x produced by the current segmentation model. The category distribution vector $\rho_x \in \mathbb{R}^{1\times C}$ for x can be calculated with R_x

$$\rho_x = \left[\frac{|R_x == 1|}{H\times W}, \frac{|R_x == 2|}{H\times W}, \ldots, \frac{|R_x == C|}{H\times W}\right] \tag{7}$$

where H, W, and C denote the height, width, and class number of the source image, respectively. We measure the similarity of x_t and x_s by cosine value of the angle between ρ_{x_t} and ρ_{x_s}

$$s = \frac{\rho_{x_\text{t}} \cdot \rho_{x_\text{s}}}{||\rho_{x_\text{t}}||_2 ||\rho_{x_\text{s}}||_2} \tag{8}$$

where "$\cdot$" is the inner product and "$||\,||_2$" means 2-norm. We need to match the images in the source and target datasets one by one. In fact, this is a matching procedure between two sets of unequal sizes, and we devised a strategy based on the idea of Hungarian algorithm [36].

The detailed matching process is shown in Algorithm 1. It is worth noting that Algorithm 1 is a simplified matching case when target dataset size n is smaller than the source dataset size m, and thus, we use the target dataset to match the source dataset. We need to repeat it and remove the matched images in the source dataset after each match until m is smaller than n, and then, we use the source dataset to match the target dataset. In this way, all images in the bigger dataset can be matched.

III. Experiments and Analysis

A. Datasets

We use three datasets to evaluate our approaches. The first two datasets are chosen from the ISPRS 2-D semantic segmentation benchmark dataset [37], i.e., the Potsdam and Vaihingen datasets, respectively. The Potsdam dataset is

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Algorithm 1 Category Similarity Matching in CSMS

Input: The source dataset size m, the target dataset size n, and the similarity between the images in two domains one by one $sim \in \mathbb{R}^{n \times m}$;

Output: the index records of matching results $s2t \in \mathbb{Z}^m$;

Obtain the index records of the sorted similarity from largest to smallest $ind_sim \in \mathbb{R}^{n \times m}$:$ind_sim = argsort(sim, axis = 1)$;

Initialize the index records of matching results $s2t = \{-1\}^m$;

Function match(ind_t_now):

  for $i = 1 \ldots m$:

    get the index of the $i-th$ similar source domain image $ind_s_now = ind_sim[ind_t_now, i]$;

    if $s2t[ind_s_now] = -1$:

      $s2t[ind_s_now] = ind_t_now$;

      break;

    else:

      get the target image index that has been matched to ind_s_now : $ind_t_last = s2t[ind_s_now]$;

      if $sim[ind_t_now, ind_s_now] > sim[ind_t_last, ind_s_now]$:

        $s2t[ind_s_now] = ind_t_now$;

        match(ind_t_last);

        break;

for $j = 1 \ldots n$:

  match(j);

collected from the city of Potsdam that contains 38 images of very high resolution with a GSD of 5 cm, whereas the Vaihingen dataset is collected from the city of Vaihingen that contains 33 very-high-resolution images with a GSD of 9 cm. All images in both datasets are provided with their semantic labels, including six land-cover categories: impervious surface, tree, car, building, low vegetation, and clutter/background. The Potsdam dataset contains three modes of different channel compositions: IR-R-G (three channels), R-G-B (three channels), and R-G-B-IR (four channels). And the first two kinds are used. The Vaihingen dataset contains only one mode: IR-R-G (three channels). Due to the computational memory constraints, we cropped the two datasets into the size of 512 × 512. For the Potsdam dataset, the cropping stride is 512 and we obtain 4598 images in each image mode. For the Vaihingen dataset, the cropping strides are 256 and we obtain 1696 images.

The third dataset was manually annotated on our own. This dataset was constructed using 2.4-m Gaofen-2 images obtained from Changchun and Guangzhou rural areas. As shown in Fig. 4, Changchun and Guangzhou are in north and south China, respectively. Their natural conditions, such as topography and climate, are very different, which leads to distinct scene styles in their rural areas. We label some RS images of rural areas in these two cities, including four land-cover categories: farmland, water, forest land, and residential land. Then, we crop them into the size of 512 × 512 with cropping

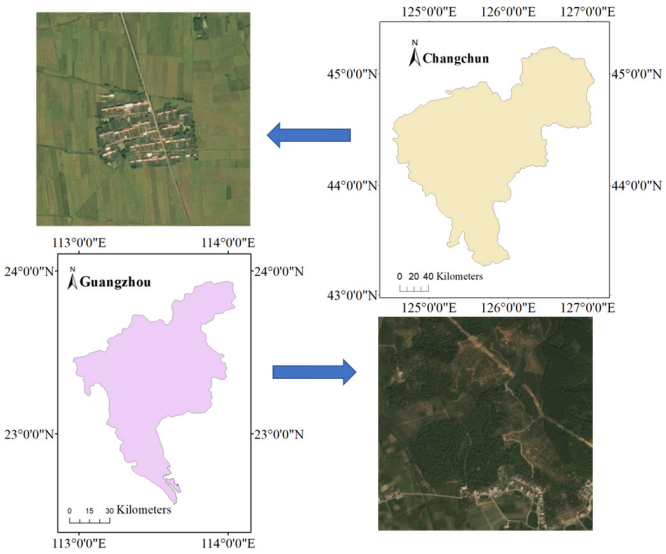


Fig. 4. Location and natural conditions of Changchun and Guangzhou.

strides of 256, and we finally obtain 986 images of rural areas in Changchun and 477 images of rural areas in Guangzhou.

B. Experimental Settings

Our method concentrates on the feature-level adaptation, i.e., although the same object may have different pixel values in the two datasets, the learned features may be consistent and robust through DA. Consequently, we design three cross-domain tasks to simulate situations that might be encountered in practical applications using the above datasets.

1) *Variation in Geographic Location and GSD*: the source domain is the Potsdam IR-R-G dataset, while the target domain is the Vaihingen IR-R-G dataset, i.e., Pot-IR-R-G → Vai-IR-R-G.
2) *Variation in Geographic Location, Channel Compositions, and GSD*: the source domain is Potsdam R-G-B dataset, while the target domain is Vaihingen IR-R-G, i.e., Pot-R-G-B → Vai-IR-R-G.
3) *Variation in Geographic Location and Natural Conditions*: the source domain is Changchun rural area RS images, and the target domain is Guangzhou rural area RS images, i.e., Changchun → Guangzhou.

For the three tasks, all the images in the source domain are used for training, while 70% of the images in the target domain are used for training and the left 30% images in the target domain are used for testing. In addition, during the training process, we also use rotation and flipping to perform random image augmentation.

In terms of experimental settings, for the first two tasks, we mainly designed the following experiments.

1) A*nalysis of the Strategy for Selecting Positive and Negative Samples*: As stated in Section II, we have two options to construct contrastive samples, and we will analyze these two strategies and explain why we utilize

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predicted labels of two domains to provide the class information.

2) *Sensitivity Analysis of the Loss Weighting Factors*: As mentioned in Section II, the loss of our model consists of three components, i.e., L_{seg}, L_{cont}, and L_{adv}, and their weights will affect the final performance of the model.
3) *Ablation Analysis*: This experiment aims to analyze the effect of each module in the model.
4) *Comparison With Other Methods*: We compare our methods with other state-of-the-art methods in semantic segmentation DA problem. The state-of-the-art methods for comparison will be described later.
5) *Analysis of the Contrastive Loss and CSMS Strategy*: We use t-distributed stochastic neighbor embedding (t-SNE) [38] to visualize the feature distribution in the target domain to see the effect of the contrastive loss and CSMS strategy.

For the third task, we design an experiment to explore the performance gained by DA pretraining, i.e., how many labeled samples we can save in the target domain by DA pretraining. In addition, we design a synthetic dataset to investigate whether DA can still work in the case where source domain and target domain have different channel sequences.

1) State-of-the-Art Methods for Comparison: We compare our method with the extensive state-of-the-art methods, including direct entropy minimization (MinEnt) [13], adapt structured output space (AdaptSegNet) [15], CLAN [16], FADA, and Tra-loss.

a) *Base:* In "Base" mode, we do not utilize any DA strategy, but only use the source dataset to train the model and test it on the target dataset. This method can clearly exhibit the domain shift problem.

b) *MinEnt:* This is a nonadversarial approach. They utilize the entropy loss to directly maximize prediction certainty in the target domain. Entropy loss can be seen as a soft-assignment version of the pseudo-label cross-entropy loss.

c) *AdaptSegNet:* Considering semantic segmentation as structured outputs that contain spatial similarities between source and target domains, AdaptSegNet adopts adversarial learning in the output space to effectively align scene layout and local context between source and target images.

d) *CLAN:* CLAN aims to address the limitation of global adversarial learning by prioritizing category-level alignment which will naturally lead to global distribution alignment. They identify the classes whose features are distributed differently between two domains and increase the weight of the adversarial loss during training.

e) *FADA:* FADA adopts a fine-grained domain discriminator that not only plays as a domain distinguisher but also differentiates domains at the class level.

f) *Base+*:* "Base+*" implies adding other branches or training strategies to the semantic segmentation network for DA, e.g., "Base + ADRS" means adding the ADRS branch.

g) *Tra-loss:* Traditional contrastive loss like Info-NCE [33] and supervised contrastive loss [32] use the sum of negative sample similarities to calculate the denominator, while we calculate the average value of them. In our experimental setting, this method differs from "Base + ADRS" only in the calculation formulation of contrastive loss. And we will use the following calculation formulas to compute the contrastive loss instead of (3):

$$\mathcal{L}_{\text{cont}}^{i_t} = \frac{1}{|F_{x_s}^{i_t+}|}\sum\nolimits_{\vec{j}\in F_{x_s}^{i_t+}} -\log\frac{\exp\left(\vec{i}_t\cdot\vec{j}/\tau\right)}{\exp\left(\vec{i}_t\cdot\vec{j}/\tau\right)+\sum_{\vec{z}\in F_{x_s}^{i_t-}}\exp\left(\vec{i}_t\cdot\vec{z}/\tau\right)} \tag{9}$$

where the notations are the same as (3).

C. Experimental Details

We use DeepLabV3+ [31] based on ResNet-50 [39] pretrained on the ImageNet [40] as the semantic segmentation network G. The feature map F produced by G has 304 channels. To avoid the GPU memory issues, in the ADRS branch, the interpolation layer resizes the feature map $F64 \times 64$ using bilinear resizing and it is normalized before computing the contrastive loss. The discriminator D consists of five convolution layers and the number of channels in each layer is {32, 64, 128, 256, 1}. The first four convolutional layers are also coupled with a leaky-ReLU layer with a fixed negative slope of 0.2.

2) Implementation Details: We employ PyTorch [41] deep learning framework in the implementations. All experiments are done on a single NVIDIA 2080TI GPU with 12-GB memory, and we set the mini-batch size as 3. Both the semantic segmentation network and the discriminator network are trained by Adam optimizer. Their initial learning rates were set as 4×10^{-4} and 3×10^{-4}, respectively, and we reduce them to one-fifth of the initial amount for every 5000 iterations during training. The total number of training iterations is 13 000, and we started matching the source and target domain images using the CSMS strategy at the 6500th iteration. The temperature τ of the contrastive loss is set to 0.5.

3) Evaluation Metrics: To quantitatively evaluate the performance of DA methods for RS image semantic segmentation, we use two kinds of measurements, i.e., intersection over union (IoU) and F1 score (F1). IoU is calculated for every class separately, and mIoU is the mean IoU of all classes. The following equation represents how to calculate IoU:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|} \tag{10}$$

where A is the set of ground-truth pixels and B is the set of predicted pixels. $\cap$ and $\cup$ denote the intersection and union operations, respectively. $|\cdot|$ denotes calculating the number of pixels in the set. Specifically, F1 score can be defined as

$$\text{F1} = \frac{2(\text{Pre}+\text{Rec})}{\text{Pre}+\text{Rec}},\ \text{Pre} = \frac{\text{tp}}{\text{tp}+\text{fp}},\ \text{Rec} = \frac{\text{tp}}{\text{tp}+\text{fn}} \tag{11}$$

where tp, fp, and fn are the true positives, false positives, and false negatives, respectively. Pre and Rec denote the precision and recall.

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TABLE I
ANALYSIS OF THE STRATEGY FOR SELECTING POSITIVE AND NEGATIVE SAMPLES ON THE TASK POT-IR-R-G → VAI-IR-R-G

Method	Option	mIoU	Mean--F1
Base+ADRS	Option (1)	37.8	51.7
	Option (2)	**39.0**	**51.8**
Base+ADRS+ADPR	Option (1)	44.1	57.3
	Option (2)	**45.7**	**60.3**

TABLE II
ANALYSIS OF THE STRATEGY FOR SELECTING POSITIVE AND NEGATIVE SAMPLES ON THE TASK POT-R-G-B → VAI-IR-R-G

Method	Option	mIoU	Mean-F1
Base+ADRS	Option (1)	29.3	42.6
	Option (2)	**37.2**	**50.1**
Base+ADRS+ADPR	Option (1)	41.6	54.5
	Option (2)	**43.2**	**56.7**

D. Analysis of the Strategy for Selecting Positive and Negative Samples

Contrastive samples are crucial in contrastive learning. In this study, there are two options providing class information for us to select positive and negative samples, that is: 1) using ground truths of the source domain and predicted labels of the target domain and 2) using predicted labels of two domains. And we test these two options on the task Pot-IR-R-G → Vai-IR-R-G and the task Pot-R-G-B → Vai-IR-R-G. The results are shown in Tables I and II. In "Base + ADRS" mode, we set the weight of the contrastive loss as 0.01; in "Base + ADRS + ADPR" mode, we set the weights of both contrastive loss and adversarial loss as 0.01. We can find that whether using contrastive loss alone or together with adversarial loss, option (2) performs better than option (1) on both Pot-IR-R-G → Vai-IR-R-G and Pot-R-G-B → Vai-IR-R-G tasks. We think that the reasons are as follows. The predicted labels are a direct reflection of the representations extracted by the current model, thus using predicted values to construct contrastive samples is softer and more consistent. Let us take an example to explain this. Given a pixel p_s from the source domain whose ground truth is low vegetation, its predicted value is tree in the ith epoch and is low vegetation in the $(i+1)$th epoch. If we use the ground truth of it, then both times we should use pixels which are predicted as low vegetation from the target domain to form sample pairs with p_s. This would obviously be misleading for the feature extraction in the target domain because the first time, the representation of p_s is closer to tree, while the second time, it is closer to low vegetation after one epoch of learning. Therefore, using predicted labels to select positive and negative samples is a more appropriate option.

TABLE III
PARAMETER ANALYSIS OF THE LOSS WEIGHTING FACTORS λ_1 AND λ_2 ON THE TASK POT-IR-R-G → VAI-IR-R-G

λ_1	λ_2	mIoU	Mean-F1
0.001	0.001	38.2	51.2
0.010	0.010	47.1	61.1
0.060	0.030	**48.5**	**63.1**
0.100	0.100	45.7	60.1

TABLE IV
PARAMETER ANALYSIS OF THE LOSS WEIGHTING FACTORS λ_1 AND λ_2 ON THE TASK POT-R-G-B → VAI-IR-R-G

λ_1	λ_2	mIoU	Mean-F1
0.001	0.001	37.6	49.6
0.010	0.010	44.2	57.7
0.030	0.040	**46.0**	**60.0**
0.100	0.100	40.7	54.5

E. Sensitivity Analysis of the Loss Weighting Factors

The loss weighting factors λ_1 and λ_2 have a significant impact on the training of the model, as there is a tradeoff among segmentation loss, contrastive loss, and adversarial loss; thus, the final transferability of the model may vary considerably with different weight settings. To analyze the sensitivity of λ_1 and λ_2, we evaluate the performance of the proposed model under different λ_1 and λ_2 on the first two tasks, i.e., Pot-IR-R-G → Vai-IR-R-G and Pot-R-G-B → Vai-IR-R-G. Tables III and IV show the results of different loss weights on the two tasks. We found that setting λ_1 and λ_2 between 0.01 and 0.10 led to better performance for both tasks. For the task Pot-IR-R-G → Vai-IR-R-G, we can see that the performance is best with an mIoU of 48.5% and a mean F1 score of 63.1% on the target test set under the setting of $\lambda_1 = 0.06$ and $\lambda_2 = 0.03$. For the task Pot-R-G-B → Vai-IR-R-G, it can be observed that the best performance is with an mIoU of 46.0% and a mean F1 score of 60.0% on the target test set under the setting of $\lambda_1 = 0.03$ and $\lambda_2 = 0.04$. Giving less power to the contrastive loss and adversarial loss, e.g., setting $\lambda_1 = \lambda_2 = 0.001$, the model performance degrades significantly, indicating the importance of the contrastive loss and the adversarial loss.

Conversely, if the two weights are set too large, the model will also perform poorly and the training process will become unstable. In addition, the task Pot-IR-R-G → Vai-IR-R-G is easier than Pot-R-G-B → Vai-IR-R-G, so the former is better performed than the latter.

F. Ablation Analyses

To verify the effectiveness of each module, we conduct ablation experiments on the task Pot-IR-R-G → Vai-IR-R-G and

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TABLE V
RESULTS OF DIFFERENT METHODS ON THE TASK POT-IR-R-G → VAI-IR-R-G

Method	Clutter		Imp suf		Car		Tree		Low veg		Building		Average	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
Base	2.7	5.3	45.1	62.1	26.2	41.5	59.4	74.5	16.9	29.0	55.4	71.3	34.3	47.3
MinEnt	15.3	26.5	62.0	76.6	14.3	25.0	58.0	73.4	11.2	20.2	63.4	77.6	37.4	49.9
AdaptSegNet	4.1	7.9	59.7	74.8	29.3	45.3	57.1	72.7	31.9	48.4	71.9	83.6	42.3	55.5
CLAN	3.1	6.0	56.5	72.2	27.7	43.4	56.8	72.5	29.1	45.1	71.2	83.2	40.7	53.7
FADA	10.6	19.2	57.1	72.7	39.8	57.0	54.2	70.3	16.8	28.8	57.2	72.8	39.3	53.5
Tra-loss	3.3	6.4	54.7	70.7	28.9	44.9	**60.0**	**75.0**	16.7	28.7	64.5	78.4	38.0	50.7
Ours (Base+ADRS)	4.6	8.9	55.1	71.0	26.8	42.3	59.9	74.9	19.8	33.1	67.7	80.7	39.0	51.8
Ours (Base+ADPR)	5.8	11.0	63.0	77.3	38.3	55.4	57.9	73.3	29.6	45.6	71.6	83.5	44.4	57.7
Ours (Base+ADRS+ADPR)	8.2	15.2	64.0	78.1	39.4	56.6	56.0	71.8	**41.9**	**59.1**	72.3	83.9	47.0	60.8
Ours (Base+ADRS+ADPR+CSMS)	**19.6**	**32.8**	**65.0**	**78.8**	**39.6**	**56.7**	54.8	70.8	36.2	53.2	**76.0**	**86.4**	**48.5**	**63.1**

TABLE VI
RESULTS OF DIFFERENT METHODS ON THE TASK POT-R-G-B → VAI-IR-R-G

Method	Clutter		Imp suf		Car		Tree		Low veg		Building		Average	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1
Base	1.8	3.6	45.6	62.6	26.8	42.3	42.4	59.6	10.1	18.3	42.5	59.6	28.2	41.0
MinEnt	8.5	15.6	60.5	75.4	16.3	28.0	53.8	69.9	17.1	29.2	60.9	75.7	36.2	49.0
AdaptSegNet	3.0	5.9	56.6	72.3	31.4	47.8	56.4	72.1	18.3	30.9	71.6	83.5	39.5	52.1
CLAN	3.1	5.9	53.8	70.0	30.8	47.1	50.6	67.2	30.9	47.2	69.7	82.1	39.8	53.2
FADA	3.3	6.4	53.2	69.4	39.3	56.5	35.6	52.5	24.5	39.3	58.5	73.8	35.7	49.6
Tra-loss	2.3	4.4	52.2	68.6	26.5	41.9	54.6	70.7	11.2	20.2	60.1	75.0	34.5	46.8
Ours (Base+ADRS)	5.3	10.0	59.8	74.8	29.8	45.9	**57.5**	**73.0**	13.4	23.7	57.7	73.1	37.2	50.1
Ours (Base+ADPR)	8.5	15.5	58.9	74.1	39.1	56.2	51.1	67.7	29.4	45.4	70.4	82.6	42.9	56.9
Ours (Base+ADRS+ADPR)	6.0	11.3	60.6	75.5	**39.2**	**56.3**	54.0	70.1	34.4	51.1	73.2	84.5	44.6	58.1
Ours (Base+ADRS+ADPR+CSMS)	**10.8**	**19.4**	**62.4**	**76.9**	38.9	56.0	53.9	70.0	**35.1**	**51.9**	**74.8**	**85.6**	**46.0**	**60.0**

the task Pot-R-G-B → Vai-IR-R-G. From Tables V and VI, as well as Figs. 5 and 6, we can see that there is a serious domain shift problem on both tasks when we only use the source dataset to train the semantic segmentation network. It can also be observed that both the ADRS and ADPR branches can improve the performance of the semantic segmentation network on the target domain.

From the results of "Base + ADRS," we can deduce that the source and target domain data can be well-aligned in the representation space using only contrastive loss. However, as can be seen in Figs. 5 and 6, the object edges in the "Base + ADRS" predictions are not very accurate. This may be caused by the loss of spatial structure information in the ADRS branch, and the reason lies in two aspects. First, the loss function contrasts features between pixels and does not consider spatial structure information. Second, ADRS downsamples the feature map before calculating the contrastive loss due to GPU memory limitations, resulting in the loss of some detailed spatial information.

"Base + ADPR" achieves impressive performance in terms of evaluation metrics, but there are still some misclassified pixels in the prediction results; especially, many pixels are misclassified as clutter category. This may be caused by the confusion of pixel features in the representation space.

From the results of "Base + ADRS + ADPR," it can be observed that when contrastive loss and adversarial loss work together, they can benefit each other and gain a superior performance. Compared with "Base + ADRS," the object edges in "Base + ADRS + ADPR" prediction are more accurate. Compared with "Base + ADPR," fewer pixels are misclassified. Finally, as shown in Tables V and VI, the model's performance is further improved by the addition of the CSMS strategy.

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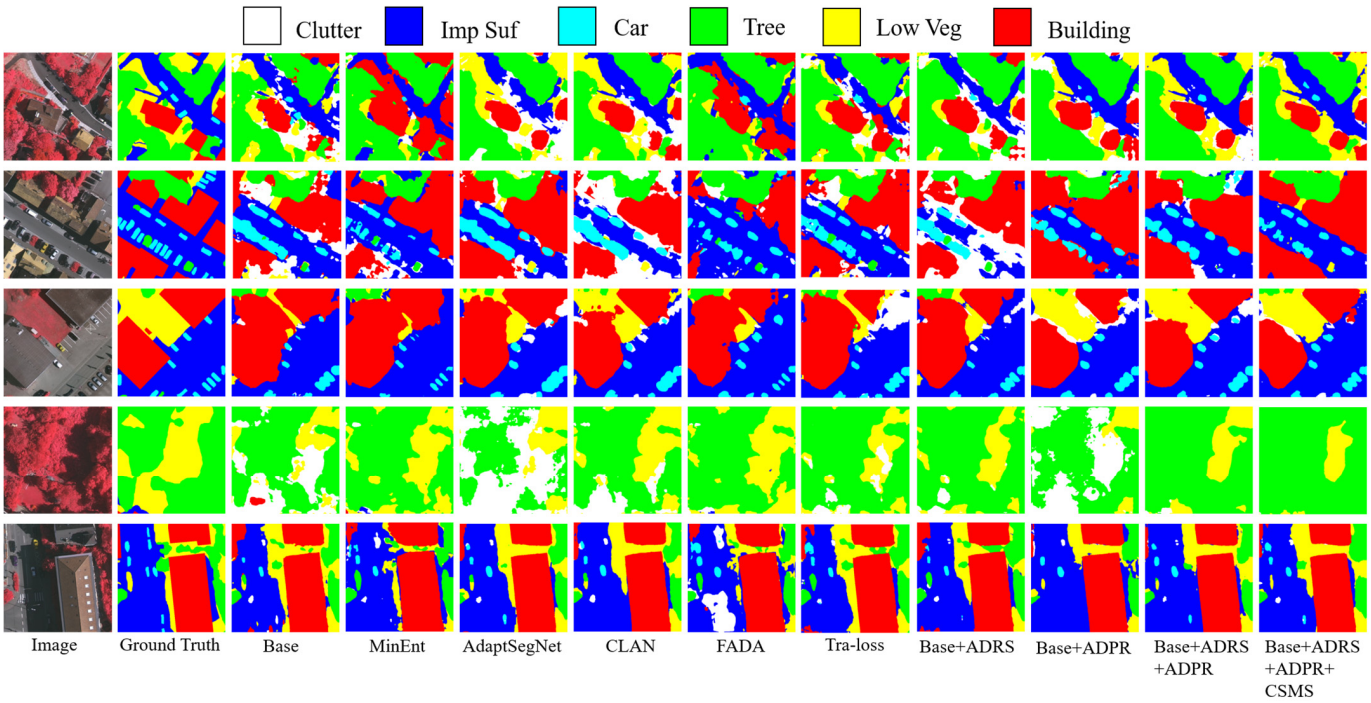


Fig. 5. Results of different methods on the task Pot-IR-R-G → Vai-IR-R-G.

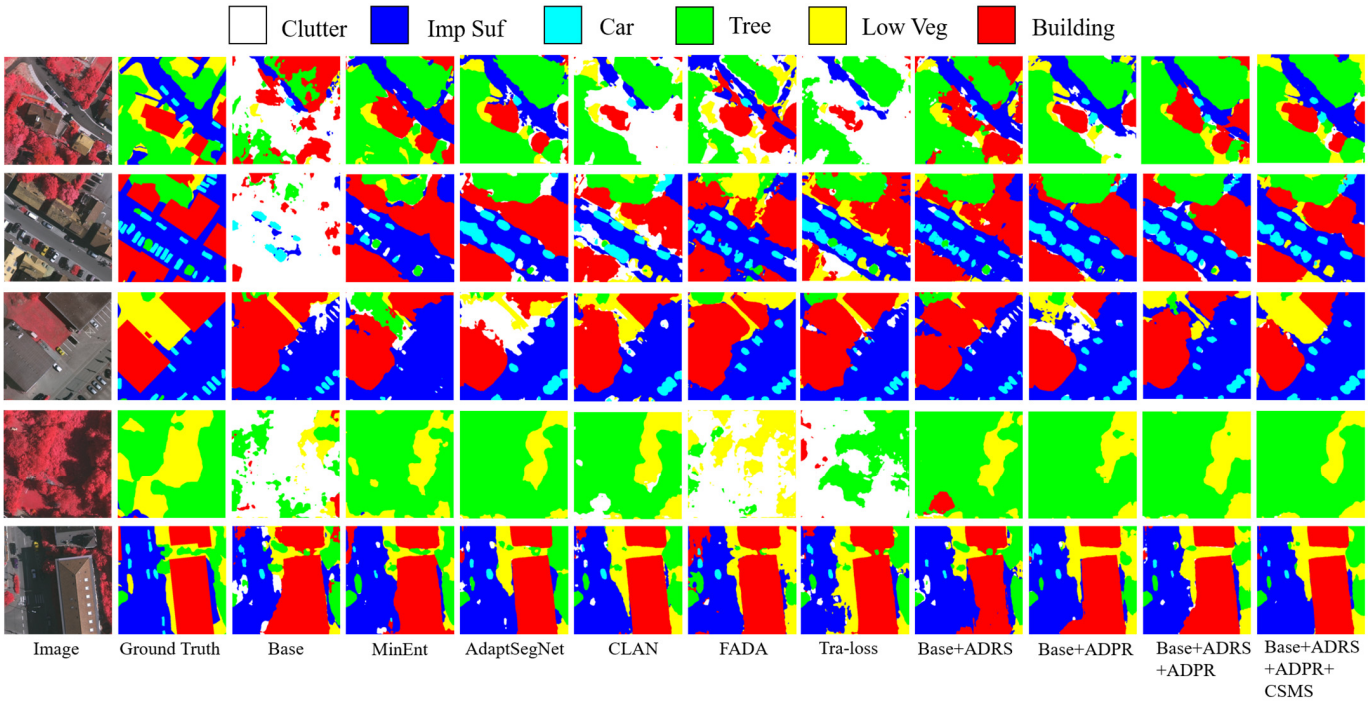


Fig. 6. Results of different methods on the task Pot-R-G-B → Vai-IR-R-G.

G. Comparison With Other Methods

In this section, we compare our methods with existing DA semantic segmentation model on the task Pot-IR-R-G → Vai-IR-R-G and Pot-R-G-B → Vai-IR-R-G. For a fair comparison, all the models used for comparison adopt DeepLabV3+ based on ResNet-50 pretrained on the ImageNet as the semantic segmentation network.

Tables V and VI show that our final model significantly improves the nonadaptive segmentation result by 14.2% and 17.8% in terms of mIoU on the task Pot-IR-R-G → Vai-IR-R-G and Pot-R-G-B → Vai-IR-R-G, respectively. In addition, our model is more effective on those hard classes, such as low vegetation, bringing over 10% and 4.2% improvement in mIoU compared with the best performance of other

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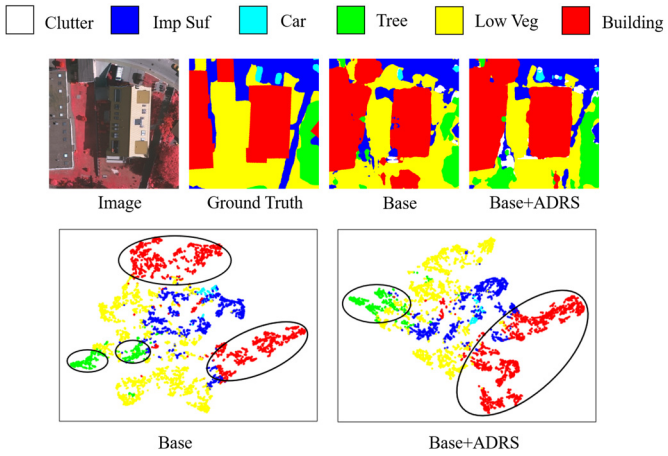


Fig. 7. t-SNE embedding results of the "Base" and "Base + ADRS" methods on the task Pot-IR-R-G → Vai-IR-R-G.

methods on the two tasks, respectively. From Figs. 5 and 6, we can see that the methods based on the adversarial learning in the output space (e.g., AdaptSegNet, CLAN, and FADA) misclassify many pixels as clutter category, but this problem does not occur with our "Base + ADRS + ADPR" model. To see the effect of contrastive loss, we can compare "Base + ADRS" with MinEnt and Tra-loss because they all use a single loss function for DA without adversarial learning, and it can be observed that our contrastive loss can achieve better results.

H. Analysis of the Contrastive Loss and CSMS Strategy

To further view the feature distribution, we randomly select a target image and map its high-dimensional features into a 2-D space with t-SNE [38]. As shown in Fig. 7, we can observe that our contrastive loss can enforce the pixels' features of the same class in the target image to be clustered together, especially for the building category and tree category in this image. To explore the role of CSMS, we test its effect on both the ADRS and ADPR branches separately. As shown in Table VII, we get more improvement by adding the CSMS strategy to "Base + ADRS" than to "Base + ADPR." This illustrates that the CSMS strategy can benefit the ADRS branch more than the ADPR branch. The CSMS strategy matches images with similar category compositions together so that more valid samples can be obtained when calculating the contrastive loss, which is important in contrastive learning.

I. Performance Gain by DA Pretraining

Collecting large-scale pixel-labeled images is time-consuming and expensive. One of the main applications of DA is to provide a good pretraining model for the target domain, so that we can retrain the model initialized by the pretrained parameter with only a small amount of labeled target domain image to achieve promising results. In this experiment, we use different numbers of target images to retrain the segmentation network with different model initialization parameters: one has its parameters pretrained by our DA method and the other does not. As shown in Fig. 8, when using only 25% of the

TABLE VII
ABLATION ANALYSIS OF CSMS STRATEGY ("+" MEANS ADDING THE CORRESPONDING MODULE TO THE MODE)

Task	Base	ADRS	ADPR	CSMS	mIoU	Mean-F1
Pot-IR-R-G → Vai-IR-R-G	+	+			39.0	51.8
	+	+		+	41.8	54.9
	+		+		44.4	57.7
	+		+	+	44.7	58.1
Pot-R-G-B → Vai-IR-R-G	+	+			37.2	50.1
	+	+		+	39.7	52.1
	+		+		42.9	56.9
	+		+	+	43.6	57.1

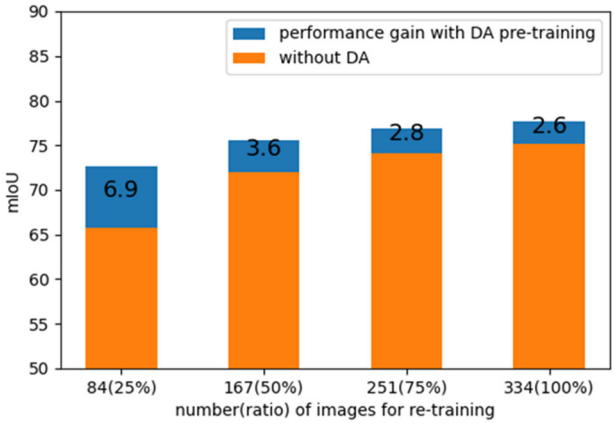


Fig. 8. Performance gain by DA pretraining.

labeled target training data, DA pretraining can improve the performance of the segmentation network by 6.9% in terms of mIoU compared to the model without DA pretraining. And when only 50% of the labeled target training data is available, the model with DA pretraining already outperforms that of the model using 100% labeled target training data without DA pretraining. The results of this experiment illustrate that the model pretrained by our DA method can achieve competitive results with a small amount of labeled target domain images.

J. Synthetic Dataset Experiments

In this section, we explore whether DA can still work in the case where source domain and target domain have different channel sequences. In specific, we use the Potsdam R-G-B dataset and change its channel sequence to generate the synthetic dataset Potsdam G-R-B, and take the former as source dataset and the latter as target dataset, i.e., "Pot-R-G-B → Pot-G-R-B." Same as other tasks, all the images in the source domain are used for training, 70% of the images in the target domain are used for training and the left 30% images in the target domain are used for testing. Intuitively, this task can be treated as a simulation of cross-spectral DA. We validate this task on all the comparing methods and ablation methods,

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TABLE VIII
RESULTS OF DIFFERENT METHODS ON THE TASK POT-R-G-B → POT-G-R-B

Method	mIoU	Mean-F1
Base	44.7	58.6
MinEnt	67.4	79.0
AdaptSegNet	72.0	82.4
CLAN	72.7	83.2
FADA	73.0	83.2
Tra-loss	69.3	80.6
Ours (Base+ADRS)	71.4	82.4
Ours (Base+ADPR)	73.6	83.9
Ours (Base+ADRS+ADPR)	75.8	85.3
Ours (Base+ADRS+ADPR+CSMS)	**76.1**	**85.5**

and the results are shown in Table VIII. The mIoU of "Base" achieves 44.7%, implying that DNNs can generalize to the dataset with the same channels but different channel sequences to some extent without DA. All the DA methods achieve more than 65.0% mIoU, which suggests that DA still works in this task. Among all the comparing baseline models, FADA shows the best performance, and our method ("Base + ADRS + ADPR + CSMS") improves by 3.1% in terms of mIoU compared to it. "Base + ADRS" mode outperforms Tra-loss, which shows the superiority of our pixel-contrastive loss. In ablation study, we can see that by combining the ADRS branch and ADPR branch, the mIoU of "Base + ADRS + ADPR" mode achieves 75.8%. And with the addition of the CSMS strategy, the model's performance is further improved. Combining the results of Pot-R-G-B → Vai-IR-R-G, we can know that our method can alleviate the cross-spectral domain shift problem to some extent.

IV. CONCLUSION

In this work, we propose a model for RS image semantic segmentation DA by integrating contrastive learning and adversarial learning to align source and target domains in both representation space and spatial layout. In addition, a training strategy is proposed to help the model work better. From the experimental results, we can draw the following conclusion.

1) The proposed pixelwise contrastive loss can align source and target domains in the representation space directly. In addition, reasonable positive and negative samples selection, e.g., using predicted labels to construct contrastive samples, can improve the performance of contrastive learning.
2) The ADRS branch and the ADPR branch in the model can benefit each other and gain superior performance. The results show that our model significantly improves the nonadaptive segmentation result by 14.2% and 17.8% in terms of mIoU on the task Pot-IR-R-G → Vai-IR-R-G and Pot-R-G-B → Vai-IR-R-G, respectively.
3) DA pretraining can help us achieve impressive results with only a small number of labeled images on the target domain.

In the future, we plan to further improve this method from the following aspects: 1) improving the CSMS strategy to help contrastive loss to obtain more positive and negative samples, thus improving its ability for DA and 2) trying other methods to help compensate for the loss of spatial structure information in the contrastive loss.

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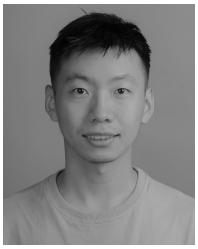
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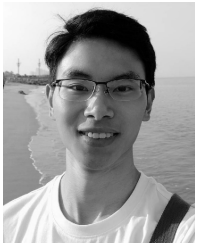
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